ModelIt!

Teacher's Guide

The Wildfire Feedback Loop

Modeling How Fire, Climate, and Ecosystems Create a Self-Reinforcing Cycle

Quick Reference

Grade Level	Grade 09L1
NGSS Standard	HS-ESS3-3, HS-LS2-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS3-3, HS-LS2-6

Create a computational simulation of a natural or designed system to test hypotheses about the functioning of natural systems. Evaluate claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students create computational simulations that model fire behavior, track changes over multiple fire cycles, and test hypotheses about how feedback loops maintain or destabilize ecosystem conditions.
Disciplinary Core Idea	**ESS3.C: Human Impacts on Earth Systems / LS2.C: Ecosystem Dynamics, Functioning, and Resilience** — Students model how human activities (fire suppression, climate change, development) alter natural fire regimes, and how complex interactions in fire-adapted ecosystems are disrupted when the natural cycle is broken.
Crosscutting Concept	**Stability and Change** — Students analyze how fire-adapted ecosystems maintain stability through periodic low-intensity disturbance, and how disruption of this cycle — through suppression, climate change, or development — pushes the system toward catastrophic, self-reinforcing change.

CAST Testing Alignment

ANALYZE:

A national forest has not burned in 80 years due to aggressive fire suppression. Using your model, explain why this forest is now at HIGHER risk of catastrophic fire than one that experienced periodic low-intensity burns. Identify every component affected by the long suppression period.

EVALUATE:

A forest manager proposes two strategies: (A) Continue fire suppression and invest in firefighting technology, or (B) Implement a prescribed burn program that intentionally sets low-intensity fires. Using evidence from your model, evaluate which strategy is more likely to prevent catastrophic wildfire in the long term.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Fire has been part of Earth's ecosystems for 400 million years — longer than flowering plants, longer than dinosaurs. Many ecosystems didn't just survive fire; they EVOLVED to need it. Pine cones that only open in extreme heat, grasses that regrow from underground roots, wildflowers that germinate in ash — fire is nature's reset button. But humans have disrupted this cycle in two devastating ways: a century of fire suppression that let fuel accumulate to explosive levels, and climate change that's drying out landscapes faster than they can recover.

The Natural Fire Cycle

In a healthy fire-adapted ecosystem, low-intensity surface fires sweep through every 5-30 years, burning dead leaves, fallen branches, and small shrubs while leaving mature trees largely unharmed. These fires recycle nutrients back to the soil, create openings for new growth, kill disease-causing organisms, and prevent fuel from accumulating to dangerous levels. Many species depend on fire: longleaf pine forests, tallgrass prairies, and Australian eucalyptus woodlands all evolved with regular burning. Without fire, these ecosystems slowly degrade.

Fire Suppression: A Century-Long Mistake

After the Great Fires of 1910 burned 3 million acres and killed 85 firefighters in the American West, the U.S. adopted a policy of suppressing ALL wildfires as quickly as possible. For a century, this seemed to work — fire acreage dropped dramatically. But fuel loads skyrocketed. Forests that naturally burned every 10-15 years accumulated 50-100 years of dead wood, brush, and dense undergrowth. What would have been a mild surface fire in 1920 is now a catastrophic crown fire in 2025 because there's 10 times more fuel.

Climate Change: Adding Fuel to the Fire

Climate change is supercharging the wildfire problem through multiple mechanisms: higher average temperatures dry out vegetation faster, snowpack melts earlier reducing summer soil moisture, drought periods are longer and more severe, and the fire season has extended by 80+ days in the western U.S. since 1970. The result: fire weather conditions that would have been extreme are now routine. The 2020 U.S. fire season burned 10.1 million acres. The 2023 Canadian wildfires burned 45 million acres — an area larger than Florida.

The Positive Feedback Loop

Here's where the science gets frightening: wildfire itself creates conditions for worse future fires. A catastrophic fire destroys the tree canopy that shades the ground → without shade, soil temperatures rise and moisture evaporates → drier soil can't support rapid revegetation → the burned area accumulates dead, dry material as fuel → the next fire has abundant dry fuel and burns even hotter. Meanwhile, massive fires release stored carbon as CO_2 → more CO_2 accelerates climate change → climate change creates hotter, drier conditions → which make fires worse. The loop feeds itself at every level.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify wind speed, fuel load, fire intensity, soil moisture, and vegetation recovery as the key components that determine wildfire behavior and ecosystem recovery.

E **Establish Relationships**

Students determine that wind speed drives fire intensity when fuel is available, fire intensity reduces soil moisture and destroys vegetation, and slow vegetation recovery leaves fuel and dry conditions for the next fire — forming a positive feedback loop.

V **Visualize & Model**

Students build a computational model showing the positive feedback loop where each fire cycle creates conditions for more intense future fires, and where disruption points could break the cycle.

E **Evaluate Outcomes**

Students run healthy fire, catastrophic fire, and post-fire recovery scenarios to test their model and discover whether the ecosystem recovers or enters a self-reinforcing destructive cycle.

R **Revise & Extend**

Students add carbon release, fire weather index, or pyrocumulonimbus events to model more realistic wildfire dynamics and climate feedback.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"In 2023, Canadian wildfires burned an area larger than Florida and turned New York City skies orange. In 2025, the LA fires destroyed over 12,000 structures. These aren't random events — they're part of a cycle that's feeding itself."*

Watch For

Show the iconic orange sky photos from NYC and satellite imagery of Canadian fires. Let the scale sink in.

Slide 2: Learning Objectives

3 min

Say: *"Today we're modeling why wildfires are getting worse — and discovering that the solution isn't fighting fire harder. It's understanding the loop."*

Watch For

Have students read objectives. Pre-teach 'fuel load' and 'positive feedback loop.' Ask who's been affected by wildfire smoke.

Slide 3: Big Question

3 min

Say: *"Here's the paradox: we've spent a century putting out fires, and fires have gotten WORSE. How is that possible?"*

Watch For

Quick-write: Students brainstorm why fighting fire might make fire worse. Pair-share. Introduce the concept of unintended consequences.

Slide 4: LEVER Framework

2 min

Say: *"We're going to model the wildfire feedback loop and find out: is there a way to break the cycle, or is it already running away from us?"*

Watch For

Preview LEVER steps. Emphasize that this model includes a FEEDBACK LOOP — the output becomes the next input.

Slide 5: Activity 1: Components

8 min

Say: *"Wind is the one thing we absolutely cannot control in a wildfire. Everything else is part of the system that feeds or fights the fire. Sort them."*

Watch For

Guide sorting: wind speed is external. Fuel load, fire intensity, soil moisture, and vegetation recovery are internal system components. Discuss controllability.

Slide 6: Activity 2: Connections

8 min

Say: *"Trace the loop: fire burns → vegetation gone → soil dries → more fuel for next fire → bigger fire. Where could we break this cycle?"*

Watch For

Help students map the complete feedback loop. Critical moment: identify where the loop can be broken (fuel reduction, prescribed burns).

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's run three futures: a healthy forest with natural fire, a catastrophic fire, and the aftermath. Does the ecosystem bounce back or spiral down?"*

Watch For

Students predict outcomes. Run scenarios over multiple fire cycles. Key discovery: the feedback loop — each fire makes the next one worse.

Slide 8: Discoveries

5 min

Say: *"Fire isn't the enemy. The ABSENCE of fire — for a century — created an enemy we may not be able to stop. The solution isn't more firefighting. It's more FIRE — the right kind, at the right time."*

Watch For

Discuss the paradox: fighting fire made things worse. Indigenous fire management used controlled burns for thousands of years. Connect to climate change amplification.

Slide 9: STEM Challenge

12 min

Say: *"A mountain town of 15,000 people borders a forest that hasn't burned in 80 years. The fuel is piled to the ceiling. Design a plan that saves the town AND the forest."*

Watch For

Groups design comprehensive wildfire resilience plans. Must include prescribed burns, defensible space, early warning, and ecological restoration. Present strategies.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Wind Speed
- **INTERNAL:** Fuel Load, Fire Intensity, Soil Moisture, Vegetation Recovery

Relationships

- **Wind Speed to Fire Intensity:** POSITIVE (+) — Stronger winds deliver more oxygen to the fire front, increase flame length and spread rate, carry embers to start spot fires ahead of the main front, and can push a surface fire into tree crowns — transforming a manageable fire into an unstoppable inferno.
- **Fuel Load to Fire Intensity:** POSITIVE (+) — More available fuel — dry wood, leaves, brush, and dead trees — provides more combustible material for the fire to consume, releasing more energy and sustaining higher temperatures and faster spread rates.
- **Fire Intensity to Soil Moisture:** NEGATIVE (-) — Intense fires destroy the vegetation canopy that shades the ground, bake moisture out of the upper soil layers, and can create hydrophobic (water-repellent) compounds in the soil that prevent rainfall from absorbing, leading to rapid drying and erosion.
- **Soil Moisture to Vegetation Recovery:** POSITIVE (+) — Higher soil moisture supports faster seed germination, root development, and plant growth after a fire. Without adequate moisture, regeneration stalls and the burned landscape remains barren, exposed to erosion and further drying.

Key Discoveries

- Low-intensity fires are natural and beneficial — they clear fuel buildup, recycle nutrients, and promote biodiversity, keeping forests healthy and reducing the risk of catastrophic fires
- A positive feedback loop exists: intense fire destroys vegetation → less vegetation means less water retention → drier soil creates drier fuel → next fire burns hotter → even more vegetation destroyed
- Decades of fire suppression have created unnaturally high fuel loads in many forests, turning what should be low-intensity surface fires into catastrophic crown fires that destroy entire ecosystems
- Climate change is accelerating the feedback loop by raising temperatures, extending drought seasons, and reducing soil moisture — creating fire conditions that exceed the ecosystem's ability to recover

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"All wildfire is destructive and should be prevented"	Many ecosystems REQUIRE fire to remain healthy. Longleaf pine forests, tallgrass prairies, chaparral, and Australian eucalyptus woodlands all evolved with regular fire and decline without it. Low-intensity surface fires recycle nutrients, reduce disease, promote biodiversity, and prevent catastrophic fuel accumulation. The problem isn't fire itself — it's the wrong KIND of fire at the wrong TIME, which results from decades of suppression.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Climate change is the only reason wildfires are getting worse"	Climate change is a major accelerant, but it's not the whole story. A century of fire suppression has loaded forests with 5-10 times more fuel than they would naturally carry. Development in the wildland-urban interface has put millions of homes in fire's path. And the loss of Indigenous cultural burning practices removed a management tool that maintained healthy landscapes for millennia. The wildfire crisis is the result of MULTIPLE human decisions compounding over time.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Bigger firefighting budgets will solve the wildfire problem"	The U.S. spends over $3 billion per year on wildfire suppression, and fires are still getting worse. Suppression treats the symptom (active fire) without addressing the cause (excessive fuel, dry conditions, climate change). Worse, every dollar spent on suppression is a dollar NOT spent on prevention — prescribed burns, fuel reduction, and building codes. The model shows that breaking the feedback loop BEFORE a fire starts is exponentially more effective than fighting it once it's burning.